



SERIES #1: CRACK MNC JAVA INTERVIEW

About Series #1: Crack MNC Java Interview

Who Is It For:

- This series is designed for freshers, graduates, and professionals with 1-5 years of experience.
- It focuses on FAQ questions relevant to major MNC interviews.

How to Use This Guide:

- Use this to learn how real-time interviews ask questions 🎯.
- Focus on applying concepts and showcasing your project experience 💻🚀.

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DAY #2: OOPs & Core Java Keywords Interview Guide

Core Java

Keyword	Simple Explanation
 <code>this</code>	Refers to the current object. Example: <code>this.name = "Java";</code>
 <code>new</code>	Creates new objects. Example: <code>String str = new String("Hello");</code>
 <code>super</code>	Refers to the parent class. Example: <code>super.methodName();</code>
 <code>import</code>	Imports external classes or packages. Example: <code>import java.util.ArrayList;</code>
 <code>instanceof</code>	Checks if an object is an instance of a class. Example: <code>if (obj instanceof String).</code>
 <code>null</code>	Represents no value or object. Example: <code>String name = null;</code>

Top FAQ Related to Static Keyword in Java 🧑‍💻

Static Variables 🔄

- When we declare static variables, they are loaded at the time of class loading and allocate memory only once.

public class

Static Block 📦

- Generally used to initialize the static variables of a class.
- It is executed only once when the class is loaded by the JVM.

Static Class 🚫 (In an interview, you might be asked: Can we declare a class as static?)

- ❌ There is no concept of a static class in Java.
- A class cannot be declared as static in Java.
- However, you can use static variables, methods, and blocks to modify the behavior of the class's members.

Where to Use Static Block in Your Project 📌

- **Creating Constants:** Initialize values like API keys or configuration settings.
- **Utility Methods:** Set up reusable utilities during class loading.
- **Factory Methods:** Perform one-time setup for object creation or initialization.
- **Tracking Instances:** Keep track of the number of instances created for monitoring or debugging.

Advantages of Using Static Block in Your Project ✅

- **Access Without Object:** Static members can be accessed directly using the class name, without creating an object.

Example: `ClassName.staticVariable;`

- **Memory Efficiency:** Shared variables reduce memory usage when multiple instances of the class are created, as they are stored only once.

Disadvantages of Using Static ❌

1. Memory Issue 🧠

- Static variables are not part of garbage collection.
- They are loaded at the time of class loading in one shot.
- Overusing static variables should be avoided.
- If a static variable is no longer needed in the application, it still remains in memory.
- Static variables are not eligible for garbage collection, leading to potential memory issues.

2. Thread Safety 🔒

- As static variables are shared, they can cause issues in a multithreaded environment.
- Extra care is required to ensure proper synchronization when using static variables across threads.

Top FAQ Related to Static Keyword in Java 🧩

Static Blocks vs. Static Methods ⚖️

- Static blocks are **used to initialize** static members of a class.
- Static blocks are executed when the class is **loaded by the Java Virtual Machine (JVM)**.
- They **run before** any instance of the class is created.
- This makes them useful for tasks that need to be performed **before any instance of the class is used**.

In Simple we can say :

- Static blocks are **useful for initializing static variables** and performing one-time initialization tasks for a class.
- They ensure that the **initialization is performed only once**, even if multiple instances of the class are created.

Top FAQ Related to final Keyword in Java 🧩

What is final keyword 💡

- The final keyword in Java is used **to restrict the behavior of classes**, methods, and variables.

final Variable 🔒

- To make a variable constant
- Once a variable is declared final, **it cannot be reassigned a new value**.

final Method 🚫

- To prevent a method from being overridden: When a method is declared final, it cannot be overridden by a subclass.

final Class 🏠

- To prevent a class from being subclassed: When a class is declared final, it cannot be subclassed by any other class.
- To ensure immutability in a class: By declaring a class final, you can ensure that it cannot be extended and its behavior cannot be modified.

Case Study & Tricky Questions on final Keyword 🧠?

Behavior of final Keyword with Overloading 🔁

- The final keyword has **no effect on overloading**.
- You can have multiple methods with the same name and different parameters, regardless of whether they are declared final or not.

Behavior of final Keyword with Overriding 🔁

- When a method is declared final in a superclass, **it cannot be overridden in any of its subclasses**.
- When a class is declared final, all of its methods are effectively final and cannot be overridden in any subclass.

What We Have Covered So Far 📖 ✨

- **Static Keyword** 🧑🏫

- Static methods can be **overloaded**, but they are **hidden**, not overridden.

- **Final Keyword** 🚫

- Final methods **cannot be overridden**.
- Final classes **cannot be subclassed**.

Your Task 📝

- Spend **30 minutes** exploring **code snippets** related to:
 1. **Static and final in overloading.**
 2. **Static and final in overriding.**

How It Works in Interviews 🤔 💬

We asked our **community members** about their interview experiences, and here's what they shared:

- 💡 **Focus Areas:** Static and final keywords.
- ❓ **Common Questions:**
 - Can static methods be overridden?
 - How does final work in overloading and overriding?
 - Real-world examples of using static and final.

Practice these concepts with code snippets to feel confident in your answers! 😊

Share Your Experience 💬 ✨

Have you attended any interviews?

✉️ **Send us your story** at support@javatechcommunity.com, and we'll share it with the community to help others! 🌟

Java Keywords Cheat Sheet 📝 ✨ #For last-minute interview brushups:

🔧 Primitive Types and Void

Keyword	Simple Explanation
 boolean	Stores true or false. Example: <code>boolean isActive = true;</code>
 byte	Smallest integer type (8-bit). Example: <code>byte age = 20;</code>
 char	Stores one character. Example: <code>char grade = 'A';</code>
 short	A 16-bit integer for small numbers. Example: <code>short distance = 1000;</code>
 int	A 32-bit integer for regular whole numbers. Example: <code>int salary = 50000;</code>
 long	A 64-bit integer for larger values. Example: <code>long population = 7000000000L;</code>
 float	Stores decimal numbers (32-bit). Example: <code>float price = 5.99f;</code>
 double	Stores precise decimal numbers (64-bit). Example: <code>double percentage = 99.99;</code>
 void	Specifies no return value for methods. Example: <code>void printMessage() { System.out.println("Hello"); }</code>

Java Keywords Cheat Sheet 📝 ✨ #For last-minute interview brushups:

🔄 Control Flow

Keyword	Simple Explanation
👉 if	Executes code if a condition is true. Example: <code>if (a > b) { }.</code>
🔄 else	Executes if the if condition is false. Example: <code>else { System.out.println("No"); }.</code>
⚠️ try	Tests a block of code for errors. Example: <code>try { int x = 5 / 0; }.</code>
🛡️ catch	Handles exceptions from a try block. Example: <code>catch (Exception e) { }.</code>
✅ finally	Always executes after try-catch. Example: <code>finally { System.out.println("Done"); }.</code>
🔄 do	Executes code at least once before checking the condition. Example: <code>do { count++; } while (count < 5);.</code>
🔄 while	Repeats code while a condition is true. Example: <code>while (x < 10) { x++; }.</code>
🔄 for	Loops through a block of code. Example: <code>for (int i = 0; i < 5; i++) { System.out.println(i); }.</code>
⏭️ continue	Skips the current loop iteration. Example: <code>if (i % 2 == 0) continue;.</code>
🛑 break	Exits the loop or switch case. Example: <code>if (x == 5) break;.</code>

Java Keywords Cheat Sheet 📝 ✨ #For last-minute interview brushups:

🔒 Modifiers

Keyword	Simple Explanation
🔒 public	Makes code accessible from anywhere. Example: <code>public class Car { }</code> .
🔒 protected	Accessible in the same package or subclasses. Example: <code>protected int speed;</code> .
🔒 private	Accessible only within the class. Example: <code>private String name;</code> .
📄 abstract	Declares incomplete classes or methods. Example: <code>abstract class Shape { abstract void draw(); }</code> .
⚙️ static	For class-level variables or methods. Example: <code>static int count = 0;</code> .
🚫 final	Prevents inheritance or modification. Example: <code>final int MAX = 100;</code> .
📦 transient	Skips during serialization. Example: <code>transient String password;</code> .
🌐 volatile	Makes variables thread-safe. Example: <code>volatile int sharedData;</code> .
🔄 synchronized	Prevents multiple threads from accessing code simultaneously. Example: <code>synchronized void update() { }</code> .
🌍 native	Allows calling methods written in another language. Example: <code>native void methodName();</code> .

Java Keywords Cheat Sheet 📝 ✨ #For last-minute interview brushups:

📦 Declarations

Keyword	Simple Explanation
🏠 class	Creates a class. Example: <code>class Student { }.</code>
👉 interface	Defines abstract methods for other classes to implement. Example: <code>interface Animal { void eat(); }.</code>
📋 enum	Declares a fixed set of constants. Example: <code>enum Colors { RED, GREEN, BLUE; }.</code>
⬆️ extends	Specifies inheritance between classes. Example: <code>class Dog extends Animal { }.</code>
🔗 implements	Implements methods from an interface. Example: <code>class Cat implements Animal { }.</code>
📁 package	Groups related classes together. Example: <code>package mypackage;.</code>
⚠️ throws	Declares exceptions a method might throw. Example: <code>void readFile() throws IOException { }.</code>

Java Keywords Cheat Sheet 📝⭐ #For last-minute interview brushups:

🔧 Primitive Types and Void

⭐ Keyword	💡 Usage
 boolean	Represents a true/false value.
 byte	8-bit integer data type.
 char	16-bit Unicode character.
 short	16-bit signed integer.
 int	32-bit signed integer.
 long	64-bit signed integer.
 float	32-bit floating point.
 double	64-bit floating point.
 void	Specifies no return value.

Task for readers:

These are the most FAQ OOP basics! 🚀

- **📖 Explore More:** Dive deeper into OOP FAQs, practice, and try small code snippets for each concept 📖. We'll share detailed code snippets on the community channel soon. **For now, we want you to explore and learn!** 🚀
- **🔧 For Experienced Professionals:** Reflect on how you've implemented these concepts in your current project.
- **🎓 For College Students & Graduates:** Relate these concepts to your syllabus and try implementing small projects.

Pro Tip: Think Big, Start Small! 🌟 Let's grow together!

Note:

- This document is meant for brush-ups key concepts & Top FAQ.
- We always recommend working on projects and practicing hands-on coding to solidify your concepts. 📖🚀

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